Introduction to Probability
Second Edition

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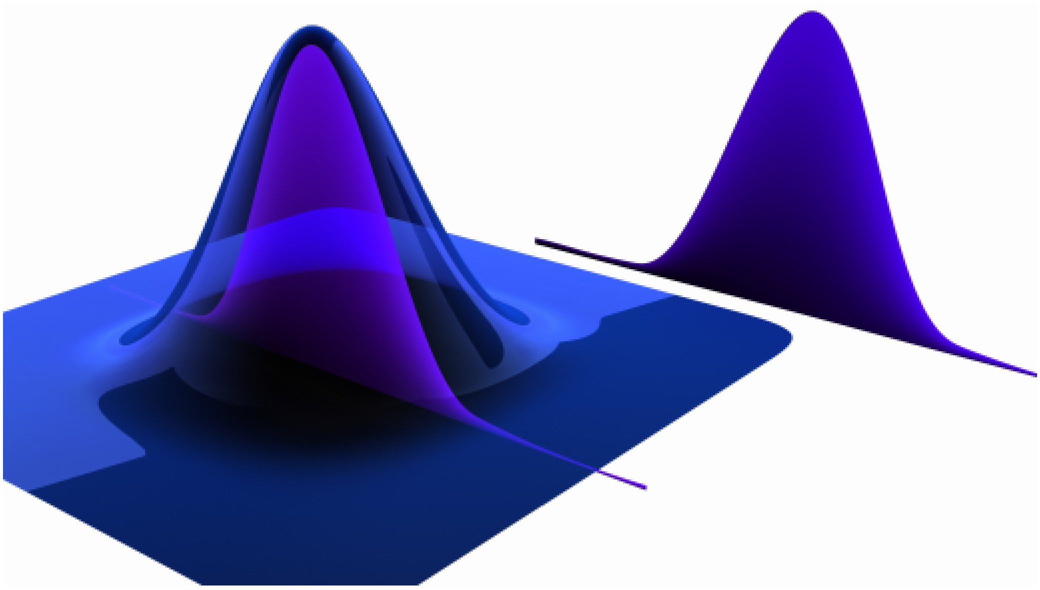

To our mothers, Steffi and Min

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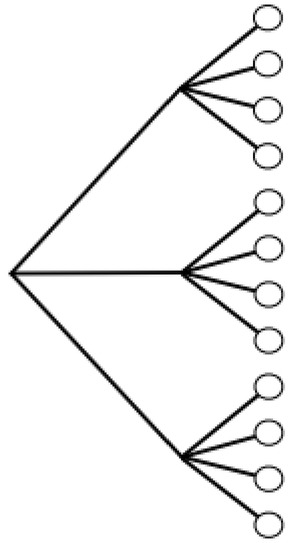

FIGURE 1.2
Tree diagram illustrating the multiplication rule. If Experiment A has 3 possible outcomes, for each of which Experiment B has 4 possible outcomes, then overall there are $3 \cdot 4 = 12$ possible outcomes.

there are 8 possibilities for third place. So by the multiplication rule, there are $10 \cdot 9 \cdot 8 = 720$ possibilities.

We did not have to consider the first place winner first. We could just as well have said that there are 10 possibilities for who got third place, then once that is fixed there are 9 possibilities for second place, and once those are both fixed there are 8 possibilities for first place. Or imagine that there are 3 platforms, which the first, second, and third place runners will stand on after the race. The platforms are gold, silver, and bronze, allocated to the first, second, and third place runners, respectively. Again there are $10 \cdot 9 \cdot 8 = 720$ possibilities for how the platforms will be occupied after the race, and there is no reason that the platforms must be considered in the order (gold, silver, bronze). □

Example 1.4.4 (Chessboard)**.** How many squares are there in an 8×8 chessboard, as in Figure 1.3? Even the name "8×8 chessboard" makes this easy: there are $8 \cdot 8 = 64$ squares on the board. The grid structure makes this clear, but we can also think of this as an example of the multiplication rule: to specify a square, we can specify which row and which column it is in. There are 8 choices of row, for each of which there are 8 choices of column.

Furthermore, we can see without doing any calculations that half the squares are white and half are black. Imagine rotating the chessboard 90 degrees clockwise. Then all the positions that had a white square now contain a black square, and vice versa, so the number of white squares must equal the number of black squares. We

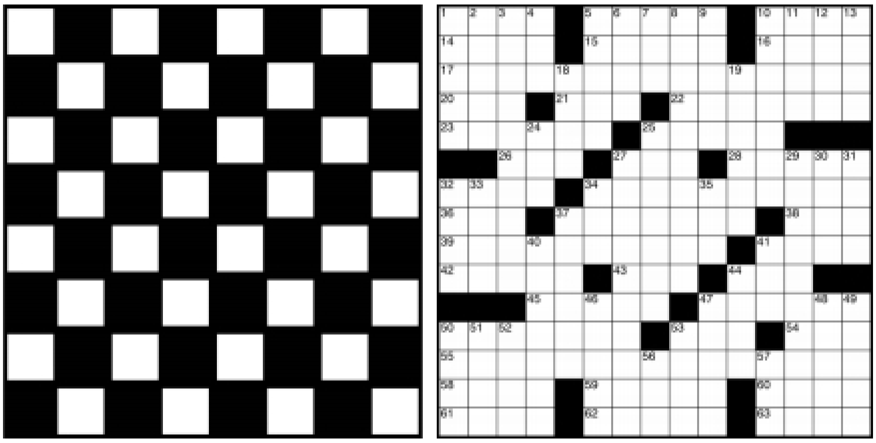

FIGURE 1.3
An 8×8 chessboard (left) and a crossword puzzle grid (right). The chessboard has $8 \cdot 8 = 64$ squares, whereas counting the number of white squares in the crossword puzzle grid requires more work.

can also count the number of white squares using the multiplication rule: in each of the 8 rows there are 4 white squares, giving a total of $8 \cdot 4 = 32$ white squares.

In contrast, it would require more effort to count the number of white squares in the crossword puzzle grid shown in Figure 1.3. The multiplication rule does not apply, since different rows sometimes have different numbers of white squares. $\square$

Example 1.4.5 (Ice cream cones)**.** Suppose you are buying an ice cream cone. You can choose whether to have a cake cone or a waffle cone, and whether to have chocolate, vanilla, or strawberry as your flavor. This decision process can be visualized with a tree diagram, as in Figure 1.4.

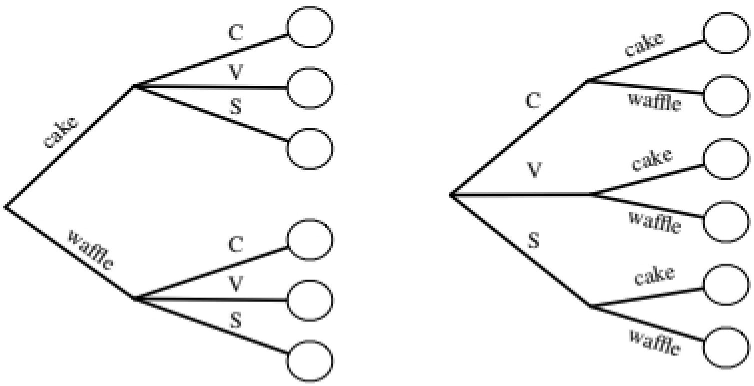

FIGURE 1.4
Tree diagram for choosing an ice cream cone. Regardless of whether the type of cone or the flavor is chosen first, there are $2 \cdot 3 = 3 \cdot 2 = 6$ possibilities.

By the multiplication rule, there are $2 \cdot 3 = 6$ possibilities. This is a very simple example, but is worth thinking through in detail as a foundation for thinking about and visualizing more complicated examples. Soon we will encounter examples where drawing the tree in a legible size would take up more space than exists in the known universe, yet where conceptually we can still think in terms of the ice cream example. Some things to note:

1. It doesn't matter whether you choose the type of cone first ("I'd like a waffle cone with chocolate ice cream") or the flavor first ("I'd like chocolate ice cream on a waffle cone"). Either way, there are $2 \cdot 3 = 3 \cdot 2 = 6$ possibilities.

2. It doesn't matter whether the same flavors are available on a cake cone as on a waffle cone. What matters is that there are exactly 3 flavor choices for each cone choice. If for some strange reason it were forbidden to have chocolate ice cream on a waffle cone, with no substitute flavor available (aside from vanilla and strawberry), there would be $3 + 2 = 5$ possibilities and the multiplication rule wouldn't apply. In larger examples, such complications could make counting the number of possibilities vastly more difficult.

Now suppose you buy *two* ice cream cones on a certain day, one in the afternoon and the other in the evening. Write, for example, (cakeC, waffleV) to mean a cake cone with chocolate in the afternoon, followed by a waffle cone with vanilla in the evening. By the multiplication rule, there are $6^2 = 36$ possibilities in your delicious compound experiment.

But what if you're only interested in what kinds of ice cream cones you had that day, not the order in which you had them, so you don't want to distinguish, for example, between (cakeC, waffleV) and (waffleV, cakeC)? Are there now $36/2 = 18$ possibilities? No, since possibilities like (cakeC, cakeC) were already only listed once each. There are $6 \cdot 5 = 30$ ordered possibilities (x, y) with $x \neq y$, which turn into 15 possibilities if we treat (x, y) as equivalent to (y, x), plus 6 possibilities of the form (x, x), giving a total of 21 possibilities. Note that if the 36 original ordered pairs (x, y) are equally likely, then the 21 possibilities here are *not* equally likely. $\square$

Example 1.4.6 (Subsets)**.** A set with n elements has 2^n subsets, including the empty set $\emptyset$ and the set itself. This follows from the multiplication rule since for each element, we can choose whether to include it or exclude it. For example, the set $\{1, 2, 3\}$ has the 8 subsets $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$. This result explains why in Example 1.2.3 there are 2^{52} events that can be defined. $\square$

We can use the multiplication rule to arrive at formulas for sampling with and without replacement. Many experiments in probability and statistics can be interpreted in one of these two contexts, so it is appealing that both formulas follow directly from the same basic counting principle.

Theorem 1.4.7 (Sampling with replacement)**.** Consider n objects and making k choices from them, one at a time *with replacement* (i.e., choosing a certain object does not preclude it from being chosen again). Then there are n^k possible outcomes

(where order matters, in the sense that, e.g., choosing object 3 and then object 7 is counted as a different outcome than choosing object 7 and then object 3.)

For example, imagine a jar with n balls, labeled from 1 to n. We sample balls one at a time with replacement, meaning that each time a ball is chosen, it is returned to the jar. Each sampled ball is a sub-experiment with n possible outcomes, and there are k sub-experiments. Thus, by the multiplication rule there are n^k ways to obtain a sample of size k.

Theorem 1.4.8 (Sampling without replacement)**.** Consider n objects and making k choices from them, one at a time *without replacement* (i.e., choosing a certain object precludes it from being chosen again). Then there are $n(n-1)\cdots(n-k+1)$ possible outcomes for $1 \leq k \leq n$, and 0 possibilities for $k > n$ (where order matters). By convention, $n(n-1)\cdots(n-k+1) = n$ for $k = 1$.

This result also follows directly from the multiplication rule: each sampled ball is again a sub-experiment, and the number of possible outcomes decreases by 1 each time. Note that for sampling k out of n objects without replacement, we need $k \leq n$, whereas in sampling with replacement the objects are inexhaustible.

Example 1.4.9 (Permutations and factorials)**.** A *permutation* of $1, 2, \ldots, n$ is an arrangement of them in some order, e.g., $3, 5, 1, 2, 4$ is a permutation of $1, 2, 3, 4, 5$. By Theorem 1.4.8 with $k = n$, there are $n!$ permutations of $1, 2, \ldots, n$. For example, there are $n!$ ways in which n people can line up for ice cream. (Recall that $n!$ is $n(n-1)(n-2)\cdots 1$ for any positive integer n, and $0! = 1$.) $\square$

Theorems 1.4.7 and 1.4.8 are theorems about *counting*, but when the naive definition applies, we can use them to calculate *probabilities*. This brings us to our next example, a famous problem in probability called the *birthday problem*. The solution incorporates both sampling with replacement and sampling without replacement.

Example 1.4.10 (Birthday problem)**.** There are k people in a room. Assume each person's birthday is equally likely to be any of the 365 days of the year (we exclude February 29), and that people's birthdays are independent (we will define *independence* formally later, but intuitively it means that knowing some people's birthdays gives us no information about other people's birthdays; this would not hold if, e.g., we knew that two of the people were twins). What is the probability that at least one pair of people in the group have the same birthday?

Solution:

There are 365^k ways to assign birthdays to the people in the room, since we can imagine the 365 days of the year being sampled k times, with replacement. By assumption, all of these possibilities are equally likely, so the naive definition of probability applies.

Used directly, the naive definition says we just need to count the number of ways to assign birthdays to k people such that there are two people who share a birthday.

Therefore

$$E(Z^{2n}) = \frac{(2n)!}{2^n \cdot n!},$$

and the odd moments of Z are equal to 0, which must be true due to the symmetry of the standard Normal. From the story proof about partnerships in Example 1.5.4, we know that $\frac{(2n)!}{2^n \cdot n!}$ is the product of the odd numbers from 1 through $2n-1$, so

$$E(Z^2) = 1, E(Z^4) = 1 \cdot 3, E(Z^6) = 1 \cdot 3 \cdot 5, \dots.$$

This result also shows that the kurtosis of a Normal r.v. is 0. For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\text{Kurt}(X) = E\left(\frac{X-\mu}{\sigma}\right)^4 - 3 = E(Z^4) - 3 = 3 - 3 = 0. \qquad \square$$

Example 6.5.3 (Log-Normal moments)**.** Now let's consider the Log-Normal distribution. We say that Y is *Log-Normal* with parameters μ and σ^2, denoted by $Y \sim \mathcal{LN}(\mu, \sigma^2)$, if $Y = e^X$ where $X \sim \mathcal{N}(\mu, \sigma^2)$.

☣ **6.5.4.** Log-Normal does not mean "log of a Normal", since a Normal can be negative. Rather, Log-Normal means "log *is* Normal". It is important to distinguish between the mean and variance of the Log-Normal and the mean and variance of the underlying Normal. Here we are defining μ and σ^2 to be the mean and variance of the underlying Normal, which is the most common convention.

Interestingly, the Log-Normal MGF does not exist, since $E(e^{tY})$ is infinite for all $t > 0$. Consider the case where $Y = e^Z$ for $Z \sim \mathcal{N}(0,1)$; by LOTUS,

$$E(e^{tY}) = E(e^{te^Z}) = \int_{-\infty}^{\infty} e^{te^z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{te^z - z^2/2} dz.$$

For any $t > 0$, $te^z - z^2/2$ goes to infinity as z grows, so the above integral diverges. Since $E(e^{tY})$ is not finite on an open interval around 0, the MGF of Y does not exist. The same reasoning holds for a general Log-Normal distribution.

However, even though the Log-Normal MGF does not exist, we can still obtain all the moments of the Log-Normal, using the MGF of the *Normal*. For $Y = e^X$ with $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$E(Y^n) = E(e^{nX}) = M_X(n) = e^{n\mu + \frac{1}{2}n^2\sigma^2}.$$

In other words, the nth moment of the Log-Normal is the MGF of the Normal evaluated at $t = n$. Letting

$$m = E(Y) = e^{\mu + \frac{1}{2}\sigma^2},$$

we have, after some algebra,

$$\text{Var}(Y) = E(Y^2) - m^2 = m^2(e^{\sigma^2} - 1).$$

All Log-Normal distributions are right-skewed. For example, Figure 6.2 shows a

Log-Normal PDF in dark, with mean 2 and variance 12. This is the distribution of e^X for $X \sim \mathcal{N}(0, 2\log 2)$, and it is clearly right-skewed. To quantify this, let us compute the skewness of the Log-Normal r.v. $Y = e^X$ for $X \sim \mathcal{N}(0, \sigma^2)$. Letting $m = E(Y) = e^{\frac{1}{2}\sigma^2}$, we have $E(Y^n) = m^{n^2}$ and $\text{Var}(Y) = m^2(m^2 - 1)$, and the third central moment is

$$\begin{aligned} E(Y - m)^3 &= E(Y^3 - 3mY^2 + 3m^2Y - m^3) \\ &= E(Y^3) - 3mE(Y^2) + 2m^3 \\ &= m^9 - 3m^5 + 2m^3. \end{aligned}$$

Thus, the skewness is

$$\text{Skew}(Y) = \frac{E(Y - m)^3}{\text{SD}^3(Y)} = \frac{m^9 - 3m^5 + 2m^3}{m^3(m^2 - 1)^{3/2}} = (m^2 + 2)\sqrt{m^2 - 1},$$

where in the last step we factored $m^6 - 3m^2 + 2 = (m^2 + 2)(m - 1)^2(m + 1)^2$. The skewness is positive since $m > 1$, and it increases very quickly as σ grows. □

In the next example, we introduce the Weibull distribution, which is one of the most widely used distributions in *survival analysis* (the study of the duration of time until an event occurs, e.g., modeling how long someone with a particular disease will live).

Example 6.5.5 (Weibull distribution)**.** As you may remember from the previous chapter, the Exponential distribution is memoryless, which makes it unrealistic for, e.g., modeling a human lifetime. Remarkably, simply raising an Exponential r.v. to a power dramatically improves the flexibility and applicability of the distribution.

Let $T = X^{1/\gamma}$, with $X \sim \text{Expo}(\lambda)$ and $\lambda, \gamma > 0$. The distribution of T is called the *Weibull* distribution, and we denote this by $T \sim \text{Wei}(\lambda, \gamma)$. This generalizes the Exponential, with the case $\gamma = 1$ reducing back to the Exponential.

Weibull distributions are widely used in biostatistics, epidemiology, and engineering; there is even an 800-page book devoted to this distribution: *The Weibull Distribution: A Handbook* by Horst Rinne [21].

The PDF of T is

$$f(t) = \gamma\lambda e^{-\lambda t^\gamma} t^{\gamma - 1}$$

for $t > 0$, as can be shown by relating the CDF of T to the CDF of X, or by using transformation results from Chapter 8. The PDF looks somewhat complicated, but often when working with a Weibull random variable we can use the strategy of first transforming it back to an Exponential.

For simplicity and concreteness, let's look at a specific Weibull distribution, letting $\lambda = 1$ and $\gamma = 1/3$.

(a) Find $P(T > s + t | T > s)$ for $s, t > 0$. Does T have the memoryless property?

(b) Find the mean and variance of T, and the nth moment $E(T^n)$ for $n = 1, 2, \ldots$.

(c) Determine whether or not the MGF of T exists.

Solution:

(a) The CDF of T is

$$P(T \le t) = P(X^3 \le t) = P(X \le t^{1/3}) = 1 - e^{-t^{1/3}},$$

for $t > 0$. So

$$P(T > s+t | T > s) = \frac{P(T > s+t)}{P(T > s)} = \frac{e^{-(s+t)^{1/3}}}{e^{-s^{1/3}}},$$

which is not the same as $P(T > t) = e^{-t^{1/3}}$. Thus, T does *not* have the memoryless property. Nor could it, since it is not Exponential.

(b) Example 6.5.1 shows that the moments of X are given by $E(X^n) = n!$. This allows us to find the moments of T without doing any additional work! Specifically,

$$E(T^n) = E(X^{3n}) = (3n)!.$$

The mean and variance of T are

$$E(T) = 3! = 6, \mathrm{Var}(T) = 6! - 6^2 = 684.$$

(c) By LOTUS,

$$E(e^{tT}) = E(e^{tX^3}) = \int_0^\infty e^{tx^3 - x} dx.$$

This integral diverges for $t > 0$ since the tx^3 term dominates over the x; more precisely, we have $tx^3 - x > x$ for all x sufficiently large (specifically, for $x > \sqrt{2/t}$), so this integral diverges by comparison with the divergent integral $\int_0^\infty e^x dx$. So the MGF of T does not exist, even though all the moments of T do exist. □

☣ **6.5.6.** Several different parameterizations of the Weibull are commonly used, e.g., we are including a scale in the Exponential and then raising to a power, but it is also common to take an Expo(1) to a power and then rescale. So care is needed when reading various references: always check which convention is being used. Our parameterization here is widely used in medical statistics, and is convenient to work with since we are just raising an Expo(λ) r.v. to a power.

6.6 Sums of independent r.v.s via MGFs

Since the MGF of a sum of independent r.v.s is just the product of the individual MGFs, we now have a new strategy for finding the distribution of a sum of independent r.v.s: multiply the individual MGFs together and see if the product is recognizable as the MGF of a named distribution. The next two examples illustrate this strategy.

Example 6.6.1 (Sum of independent Poissons)**.** Using MGFs, we can easily show that the sum of independent Poissons is Poisson. First let's find the MGF of $X \sim \text{Pois}(\lambda)$:

$$E(e^{tX}) = \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$

Now let $Y \sim \text{Pois}(\mu)$ be independent of X. The MGF of $X+Y$ is

$$E(e^{tX})E(e^{tY}) = e^{\lambda(e^t-1)} e^{\mu(e^t-1)} = e^{(\lambda+\mu)(e^t-1)},$$

which is the $\text{Pois}(\lambda+\mu)$ MGF. Since the MGF determines the distribution, we have proven that $X+Y \sim \text{Pois}(\lambda+\mu)$. Contrast this with the proof from Chapter 4 (Theorem 4.8.1), which required using the law of total probability and summing over all possible values of X. The proof using MGFs is far less tedious. □

☣ **6.6.2.** It is important that X and Y be independent in the above example. To see why, consider an extreme form of dependence: $X = Y$. In that case, $X + Y = 2X$, which can't possibly be Poisson since its value is always an even number!

Example 6.6.3 (Sum of independent Normals)**.** If we have $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ independently, then the MGF of $X_1 + X_2$ is

$$M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t) = e^{\mu_1 t + \frac{1}{2}\sigma_1^2 t^2} \cdot e^{\mu_2 t + \frac{1}{2}\sigma_2^2 t^2} = e^{(\mu_1+\mu_2)t + \frac{1}{2}(\sigma_1^2+\sigma_2^2)t^2},$$

which is the $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ MGF. Again, because the MGF determines the distribution, it must be the case that $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Thus the sum of independent Normals is Normal, and the means and variances simply add. □

Example 6.6.4 (Sum is Normal)**.** A converse to the previous example also holds: if X_1 and X_2 are independent and X_1+X_2 is Normal, then X_1 and X_2 must be Normal! This is known as *Cramér's theorem*. Proving this in full generality is difficult, but it becomes much easier if X_1 and X_2 are i.i.d. with MGF $M(t)$. Without loss of generality, we can assume $X_1 + X_2 \sim \mathcal{N}(0, 1)$, and then its MGF is

$$e^{t^2/2} = E(e^{t(X_1+X_2)}) = E(e^{tX_1})E(e^{tX_2}) = (M(t))^2,$$

so $M(t) = e^{t^2/4}$, which is the $\mathcal{N}(0, 1/2)$ MGF. Thus, $X_1, X_2 \sim \mathcal{N}(0, 1/2)$. □

In Chapter 8 we'll discuss a more general technique for finding the distribution of a sum of r.v.s, which applies when the individual MGFs don't exist, or when the product of the individual MGFs is *not* recognizable and we would like to get the PMF/PDF instead.

6.7 *Probability generating functions

In this section we discuss *probability generating functions*, which are similar to MGFs but are guaranteed to exist for nonnegative integer-valued r.v.s. First we'll use PGFs